Tide retrospective: *the base-case discharge — recovery with nothing assumed at orders* 0–2

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Abstract

The second perimeter tide from the germbij fidelity review[1] removes every base-case assumption from the superPoly-language recovery headline. The merged theorem assumed `hbase` — equal values (which the note proves are *lost*), equal gradients, equal Hessians — and a shared package matrix H. The new theorem[2] assumes none of these: superPoly-matched second-moment and monomial moment families alone force equality of every positive-order derivative tensor, and the analytic corollary (germbij Corollary 3.2, inverse direction) follows in the same freed form. Three mechanisms do the work: exact shift invariance of normalized posterior moments (the constant the note says is invisible really is invisible, as a theorem); ray coefficient uniqueness at the `HigherLaplaceDomain` level, forcing $T_1 = 0$ and $T_2 = \mathrm{qform}\, H/2$ from the fixed-ball Taylor remainder; and the parent tide's $H_1 = H_2$. With this, review findings F1 and F2 are closed.

1 Setting

The fidelity review's two deepest findings about the inverse half of germbij Theorem 3.1 were: (F1) the headline derives higher-order tensors only *conditional on* a matched 0–2 jet, with the Hessian recovery uncomposed; and (F2) the $j = 0$ case of that assumption — equal loss values at the minimum — excludes exactly the pairs $L_2 = L_1 + c$ the note's normalization argument is about. The parent tide supplied $H_1 = H_2$ from second-moment data. This tide discharges the rest and recomposes the headline. The candidate was deliberated as one tide (shift machinery has no standalone value; the parent deliberation had already assigned the tensor tie to the recomposition), with GPT contributing two statement-level refinements: the shift invariance holds *exactly at every scale*, junk-consistently where the denominator vanishes; and the second-moment data hypothesis must name specific local packages (the $k = 3$ projections), since the higher-domain family may carry different localization regions at different orders.

2 The thing formalised

For localized nondegenerate losses L_1, L_2 carrying `HigherLaplaceDomain` families at every order $k > 2$ (with matrices H_1, H_2 and symmetric derivative tensors): if the second-moment families

[1] `2026-08-10-01-29-review-germbij-new-content`; the first was the $k = 2$ covariance bridge.

[2] `smooth_positive_jet_recovery_of_superPoly_moments`, `Laplace/Multi/ShiftNormalization.lean`.

$t \mapsto \langle w_i w_j \rangle_t$ and the monomial families $t \mapsto \langle w^m \rangle_t$ (degree > 2, rescaled) of the two losses agree beyond all orders in the temperature, then

$$\forall j > 0 : \quad D^j L_1(0) = D^j L_2(0),$$

and, for losses analytic at the minimum, the germs agree modulo an additive constant. The Lean signatures conclude `iteratedFDeriv ℝ j L₁ 0 = iteratedFDeriv ℝ j L₂ 0` for `0 < j`, and $\forall^f$ `y in 𝒩 0, L₁ y - L₁ 0 = L₂ y - L₂ 0`. Compared to the merged headline, the hypotheses `hbase` and the shared `H` are gone; the symmetry debt widens by one order (`IsSymm` now demanded at $k = 2$ as well, still mathematically implied by smoothness).

3 Proof strategy

Three layers.

Shift transport. A constant shift $L \mapsto L + a$ changes nothing the packages measure: the quadratic Peano field and the coercivity field see only $L \cdot - L\,0$, and in the degree-($n{+}1$) Taylor remainder the constant is absorbed by the degree-0 diagonal term (positive-order `iteratedFDeriv` ignores constants — `fun_iteratedFDeriv_add_apply` plus `iteratedFDeriv_const_of_ne`). For the moments,

$$\int_U f\, e^{-(L+a)/q^2} = e^{-a/q^2} \int_U f\, e^{-L/q^2},$$

so the normalized moment is *exactly* invariant (`posteriorMoment_shift`): the nonzero exponential cancels by `mul_div_mul_left`, which is junk-consistent even where the denominator vanishes — no positivity side conditions anywhere.

Ray uniqueness. Along a ray $t \mapsto tz$, the loss admits two order-2 asymptotic polynomials: the diagonal Taylor coefficients (the fixed-ball remainder bound gives an $O(t^k)$ tail once tz eventually enters the ball, and $O(t^k) \subseteq o(t^2)$ for $k > 2$), and the quadratic-Peano coefficients ($L\,0$, 0, $\mathrm{qform}\, H\, z/2$). Stage-1 coefficient uniqueness at orders 1 and 2 forces $T_1 = 0$ and $T_2 = \mathrm{qform}\, H\, z/2$. A 1-multilinear map vanishing on diagonals vanishes (every `Fin 1` input is constant), giving $D^1 L(0) = 0$ for both losses; the order-2 diagonals are both $\mathrm{qform}\, H$, and polarization (`iteratedFDeriv_eq_of_diag_eq`) identifies the tensors.

Recomposition. $H_1 = H_2$ from the parent bridge; substitute. Shift L_2 by $a = L_1 0 - L_2 0$; the three base cases now hold by construction, by ray uniqueness, and by the diagonal argument respectively; the data and symmetry hypotheses transport verbatim through the shift lemmas; the merged headline applies; and un-shifting the conclusion at positive orders is one rewrite.

4 Roadblocks and resolutions

Three small ones, all Lean-mechanical. The pointwise integrand identity behind the shift factorization arrives un-beta-reduced from `integral_const_mul` (both a stray `(fun a_1 ↦ ...) w` and the structure-index loss `(fun w ↦ L w + a) w`), and `rw` refuses; a `show` with the beta-reduced statement fixes it — the same class as the catalogued Pi.add/Pi.div traps. Second, `ContDiff`'s smoothness grade now lives in `WithTop ℕ∞`, so a cast typed at ℕ∞ produces a baffling mismatch between two `Nat.cast`s; annotate the inequality at `WithTop ℕ∞`. Third, the first draft's cast chain for the per-index `ContDiffAt` was nonsense; restructuring to quantify over `i ∈ Finset.range n` made the bound available where needed. Everything else compiled as deliberated, second build green.

5 What was Mathlib, what was new

Mathlib lookup: `fun_iteratedFDeriv_add_apply`, `iteratedFDeriv_const_of_ne`, `ContDiff.add`, `Measurable.add_const`, `mul_div_mul_left`, `Finset.sum_range_succ'/sum_Ico_consecutive`, `isLittleO_pow_pow`. Seabed reuse: the stage-1 coefficient uniqueness, the GaussAbsorb ray-argument pattern (mirrored from the Peano field to the fixed-ball remainder), the J3 polarization wrapper, the parent tide's bridge, and the merged headline. New: the three shift transports, the exact moment invariance, the higher-domain ray pair, the two tensor extractions, and the two freed headlines. The mirroring of `rayExpansion_quad` from GaussAbsorb (whose proof needed only the quadratic Peano field it inherited) into `LocalQuadraticApprox` where it belongs is a small altitude correction the simplification pass had already hinted at.

6 Lessons

The note's "the additive constant is genuinely lost" has a precise formal dual: *exact* shift invariance of the normalized moments, at every scale, with no eventuality or positivity qualifiers. Proving the invisibility as an identity (rather than asymptotically) is what let the shift wrapper discharge $j = 0$ with zero analytic content. Also carried forward: `ContDiff` grades are `WithTop` $\mathbb{N}\infty$ now — casts into $\mathbb{N}\infty$ are the wrong target and the error message does not say so (catalogued for `CLAUDE.md`).

7 Follow-ups

- Tide 3 (next): the cutoff-removal lemma — C_c^∞ observable data implies the monomial data, via the loss gap — completing the note-literal premise, plus its composition with this tide's freed headline.
- Deriving `IsSymm` at order 2 from `ContDiff` (Mathlib second-derivative symmetry) would remove the last formalization debt in the hypothesis list; deliberated as avoidable API work, left as debt.
- The GaussAbsorb ray pair should eventually delegate to the versions this file states at the natural altitude[3] (dedup for the simplification pass).

[3] On `LocalQuadraticApprox` and `HigherLaplaceDomain` respectively.